

Instruction Blindness in Vision–Language–Action Policies: Diagnosis and a Low-Rank Data Cure

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Abstract

A vision–language–action (VLA) policy can score up to 97% on a manipulation benchmark while ignoring its instruction: replace the words with the string “xxx” and it emits the same actions. It succeeds by moving to the object that usually occupies that position, not by resolving the name. We call this failure mode instruction blindness and show it is a specific, measurable instance of causal confusion in imitation learning. To measure it we use a counterfactual protocol: hold the pixels, the scene, and the robot pose fixed, change one word in the instruction, and record which object the arm reaches. On this protocol every frontier policy we tested selects the correct object 0–1% of the time, the axis on which task success and language grounding diverge. We diagnose the cause three ways—the imitation-learning literature, a black-box circuit, and a null result: the pre-trained backbone resolves the named object in its language stream, but the action head routes position and never reads that binding, and no training-time regularizer on the original data repairs it. The repair is a data recipe, not an architecture. Making the instruction the only predictor of the target and protecting the pretrained grounding with a low-rank update installs grounding (13.5% $\rightarrow$ 94.5%), whereas full fine-tuning on the identical data destroys it (12.5%). The resulting policy obeys a renamed target on 200 of 200 trials, lands a median 2 mm from it, grounds seven referent-type axes in one set of weights (obey rate 75–99%, occlusion $\leq 2\%$ on every axis), and transfers to a second simulator. We release the policy, the deconfounded datasets and their generator, and the measurement code.

1 Introduction

A vision–language–action (VLA) policy can pass a manipulation benchmark without reading its instruction. Replace the instruction with the string “xxx” and a frontier policy emits the same actions (Zhou et al., 2025); it succeeds by moving to the object that usually occupies the target position, not by resolving the words. Standard benchmarks reward this, because they place the named object where the policy expects it, so a position shortcut and genuine language grounding earn the same score. On the one axis that separates them—object identity, obtained by renaming the target and keeping the scene—the published field selects the correct object 0–1% of the time (Figure 1).

We call this failure mode instruction blindness, and we treat it as a concrete instance of a known phenomenon rather than a new mystery. Causal confusion in imitation learning (de Haan et al., 2019) predicts that a policy trained by behavior cloning will latch onto whichever feature most easily predicts the expert action; when the named object reliably sits in a fixed position, that feature is position, and the instruction becomes a spurious correlate the policy is free to ignore. Instruction blindness is what causal confusion looks like when the confounded variable is the language channel of a robot policy. Naming it this way sets the agenda of the paper: locate the mechanism, and remove the confound.

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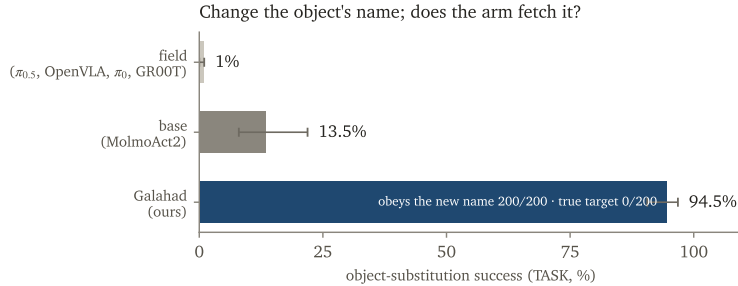


Figure 1: On the axis that separates language grounding from a position shortcut—renaming the target object while holding the scene fixed—the published field selects the correct object 0–1% of the time and the base model 13.5%. The cure of Section 6 reaches 94.5%, obeying the new name on 200 of 200 trials and never fetching the object it was trained to fetch. Intervals are Wilson 95%.

Measuring the phenomenon requires a test that a position shortcut cannot pass, and neither signal the field relies on is one. Task success is not: a policy that memorizes positions completes the task without reading the name. Vision-dependence is not: blanking the cameras collapses a position-memorizer as surely as a grounded policy. Our measurement changes one word. Holding the pixels, the scene, and the robot pose fixed, we rename the target and record where the arm goes; a grounded policy moves to the newly named object, a shortcut moves to the same place regardless of the word. This one comparison, which we call swap-obey, converts a hollow “it failed” into a positive control (“it obeyed the new name”), and it is the instrument every number in the paper is built on.

The mechanism is a severed route, not missing knowledge. The pretrained backbone already resolves the named object in its language stream—a frozen probe reads the binding, and injecting the word direction at an early layer flips the selected object 100% of the time—but the action head routes position and never reads that binding (Section 5). Because the knowledge is present and only the routing is missing, the repair is a data recipe, not an architecture. We make the instruction the only predictor of the target: every object appears as the target and as a distractor across randomized positions, so a position shortcut earns nothing and only reading the name pays. We protect the pretrained resolution with a low-rank update. On the identical data, full fine-tuning destroys grounding and returns 12.5%, while a rank-32 adapter installs it and returns 94.5%. No coupling loss, no detector, and no architecture change is added.

Contributions. This paper makes three contributions.

1. A diagnosis. We name instruction blindness, show it is field-wide with a counterfactual protocol on which every policy we tested scores 0–1% on object identity, and locate its mechanism three ways (imitation-learning theory, a black-box circuit, and a regularizer null): the object-name binding lives in the language stream, the action head routes position, and one word-direction injection flips the grounding.
2. A cure and its boundaries. Deconfounded data and low-rank protection form a causal pair—the same data under full fine-tuning destroys grounding—and we map the recipe’s edges as findings: grounding costs an object’s appearance rather than a demonstration, the vocabulary is closed, grounding trades against shortcut robustness by design, and the cure needs a base whose backbone grounding is architecturally exposed to the action head.
3. Galahad, the released artifact. The cured policy reaches 94.5% on object substitution where its base scores 13.5% and the field 0–1%, obeys a renamed target 200 of 200 times, grounds seven referent-type axes in one set of weights, and transfers to a second simulator. We release the policy, the deconfounded datasets and their generator, and the measurement code, so the diagnosis and the cure are both reproducible.

2 Related work

VLA policies and their benchmarks. A generation of open VLAs maps pixels and instructions to actions (Brohan et al., 2023; Kim et al., 2024; Black et al., 2024; Physical Intelligence, 2025; 2026; Shukor et al., 2025; NVIDIA, 2025b; Lee et al., 2025; Fang et al., 2026a), trained and scored on standard manipulation suites (Liu et al., 2023; Li et al., 2024; Nasiriany et al., 2024; Tao et al., 2024; Chen et al., 2025) in a common toolchain (Cadene et al., 2024; Chi et al., 2023). These suites measure motor competence: they place the named object where the policy expects it, so a position shortcut and grounding earn the same number. Our released policy is a low-rank fine-tune of one such base (Fang et al., 2026a); the contributions are the diagnosis, the measurement, and the recipe, not the backbone.

Instruction blindness as causal confusion. Causal confusion and simplicity/gradient-starvation bias predict that behavior cloning latches onto a sufficient easy feature and starves a harder useful one (de Haan et al., 2019; Wen et al., 2020; Pezeshki et al., 2021; Shah et al., 2020; Geirhos et al., 2020). We give this prediction a name and a measurement in robot policies: the easy feature is target position, the starved one is the instruction. Perturbation and paraphrase benchmarks expose the same gap—moving or renaming the object collapses the field to near zero on object identity (Zhou et al., 2025; Fei et al., 2025; Kim et al., 2026; Wang et al., 2026b; Emukpere et al., 2026); controlling for grasp shows target selection at chance (Yu et al., 2026); a thirty-policy sim-and-real benchmark reports best open-vocabulary instruction following at 1.67% (Chen et al., 2026); and occlusion studies confirm that fixing perception does not fix grounding (Li et al., 2026; Pumacay et al., 2024; Wang et al., 2025). Concurrent work diagnoses the vision shortcut and adds an inference-time guidance fix (Fang et al., 2026c; Lian et al., 2026; Xing et al., 2025). We differ in the readout and the fix: a positive control (the arm must go to the newly named object, not merely score a drop), and a data recipe that reaches 94.5% across seven referent-type axes rather than an architectural add-on. We also localize the mechanism, extending activation-level steering of VLAs (Häon et al., 2025) from motion to object identity.

Injecting grounding, and why we use data instead. A parallel line adds grounding through the architecture: an explicit point or box injected into the action head (Tsai et al., 2026; Yu et al., 2025), an attention-alignment object head (Jia et al., 2026), or a reasoning branch (Sun et al., 2026). We trained these on our data and cut them, installing grounding with counterfactual, role-balanced demonstrations (Pitis et al., 2020; Ameperosa et al., 2025) under a low-rank update (Hu et al., 2022) that protects the pretrained model—the gradient-isolation principle of Driess et al. (2025) by a cheaper means. Reinforcement learning sharpens what a base already samples but does not add a missing capability (Yue et al., 2025; Liu et al., 2025; Chu et al., 2025; Lin et al., 2025), so we supply the missing grounding with data, not reward. Galahad follows the template of instruction-following-from-a-base-plus-a-data-recipe (Taori et al., 2023; Wang et al., 2023).

World-action models. Models that predict a future alongside actions are an established class (Cen et al., 2025b;a; Wang et al., 2026c; Ma et al., 2026; Li et al., 2025c; Zhu et al., 2025); on our protocol the best of them obeys the instruction 20–45% of the time (Section 4), because action-conditioned foresight is a training-time regularizer, removable at inference (Fang et al., 2026b), that hallucinates success under a wrong action (Liu et al., 2026). We adopt the functional, intervention-aware definition of a world model (Wang et al., 2026a; Hansen et al., 2024); existing world-model leaderboards reward perceptual quality over instruction-correctness (Shang et al., 2026; Li et al., 2025a; Duan et al., 2025; Hu et al., 2025; NVIDIA, 2025c;a), the counterfactual-controllability gap our prediction-face metric fills (Section 8). This project began in the adversarial-policy tradition (Gleave et al., 2020; Li et al., 2025b): a policy that does not read its instruction is a hollow victim with no structure to attack, which is what turned our attention to grounding in the first place.

3 Measuring grounding: a counterfactual protocol

Every number in this paper is a behavioral counterfactual, and we state the protocol before any result because the diagnosis and the cure are only as strong as the measurement behind them.

Task success and vision-dependence do not measure grounding. A policy that memorizes where each object sits completes the task without reading the instruction, so task success certifies motor competence, not grounding. Blanking the cameras does not fix this: a position-memorizer collapses under blank cameras just as a grounded policy does, so an occlusion control (occ) rules out a state leak but not a position prior. We show this directly in Section 4: the base model passes task success on the spatial suite at 99% and on the goal suite at 100%, and both scores are shortcuts. A measurement of grounding must vary the one thing a shortcut ignores—the word.

swap-obey: rename the target and watch the arm. Our primary instrument holds the pixels, the scene, and the robot pose fixed, changes one word in the instruction, and records which object the end-effector reaches. The new name is drawn from an object actually present in the scene, so there is always a real target to go to, and we score two quantities: the success rate on the original target, which a grounded policy drives to zero, and obeyed-name, the rate at which the end-effector lands on the newly named object. obeyed-name is the load-bearing readout: it converts a hollow “the policy failed” into a positive control, “the policy obeyed the new name,” and only that direction distinguishes a policy that reads the instruction from one that rides the scene. We rotate the renamed decoy across every in-scene non-target so the protocol cannot be passed by orienting toward one familiar object.

Two disciplines keep the readout honest. We verify the distance, not just the threshold: on the cured policy the end-effector lands a median 2 mm from the newly named object while sitting a median 9 cm from the originally trained one, and no trial falls in the 4–5 cm grazing band, so a 2 cm threshold gives the same verdict as a 5 cm one. And we report a reach-probe on every failure, splitting it into reached-target (the arm reached the right object and the grasp slipped, a motor failure) and went-distractor (the arm went to the wrong object, a grounding failure), so we never record a motor limit as a grounding limit.

Three controls remove three ways to cheat. occ asks whether the policy uses vision (blank the cameras; success must collapse). swap-obey asks whether it follows the named object (rename the target; the arm must switch). nonsense asks whether it needs a name at all (replace the instruction with “xxx”; a policy that still succeeds is riding a scene prior). A wrong-pointer face confirms the policy uses no injected coordinate. Success is lift-gated and init-false: the object must leave the table by a fixed margin, so a scene in which the target begins at the goal cannot score a do-nothing success. We measured the init value of every metric and required a do-nothing policy to score near zero before trusting any rate.

One protocol, every referent type. Object identity is the sharpest axis but not the only one. The same counterfactual instantiates for each way an instruction can name a target—color, category, ordinal position, spatial relation, negation, and composition—by varying the one word that selects the referent. This is the protocol behind the field-wide result of Section 4 and the seven-axis result of Section 7. It also extends to a policy that predicts a future: we change one word and ask whether the predicted region of motion moves to the newly named object, a control the world-model evaluation literature leaves open (Section 8).

4 Instruction blindness is field-wide

We run the protocol of Section 3 across the field. The result is not a uniform wall but a graded one: policies fail to different depths, and the depth is set by how the instruction reaches the output.

The base has full motor and near-chance grounding. The base model, MolmoAct2 (Fang et al., 2026a), executes standard pick-and-place at 100% and, under object substitution, selects the correct object 13.5% of the time (measured cleanly at $n=96$, [8.0, 21.9]). The gap between 100% motor and 13.5% grounding is instruction blindness in one model: the motor works and the instruction does not drive it. This before-number also settles how much grounding the cure starts from—13.5%, not zero.

The imitation field collapses on object identity. On the object-substitution axis the published imitation field scores 0–1%: $\pi_{0.5}$ at 1%, and OpenVLA, π_0 , and GR00T at 0% (Zhou et al., 2025). We do not rest the claim on one source. Yu et al. (2026) report frontier VLAs selecting the correct object at near-chance rates after controlling for grasp success; Chen et al. (2026) run thirty policies on an independent sim-and-real benchmark and find the best open-vocabulary instruction-following score is 1.67%; and our own search for a groundable policy, the search that started this project, returned none. The newest releases repeat the pattern: an open VLA reporting 98.7% on LIBERO reproduces its published positive control on our harness (50/50), then drops to 10.0% [4.3, 21.4] under object substitution, and completes the canonical task 52% of the time with the instruction replaced by “xxx” (Unitree Robotics, 2026). The scene decides; the words are optional.

World-action models are a leaderboard, not a wall. We ran the battery on four open world-action models under a state-leak-clean protocol (blank cameras and blank proprioception both collapse success, so no model succeeds from a leaked state). Obey rates: UniVLA (Wang et al., 2026c) 20%, InternVLA-A1.5 (Ma et al., 2026) 23.3%, WorldVLA (Cen et al., 2025b) 25%, and RynnVLA-002 (Cen et al., 2025a) 45%. The best open world-action model reaches roughly half the grounding of our cured policy, and the rest sit near chance. Their prediction heads are action-conditioned and instruction-blind at the frame level, which is why grounding of a predicted future tracks the conditioning channel rather than model scale; we quantify that spectrum—text-conditioned generation grounds far better than action-conditioned world models—in Appendix A. The graded leaderboard is the same disease read at different depths: the closer the instruction sits to the output objective, the more of it survives.

5 Why imitation is instruction-blind

Instruction blindness is a severed route, not missing knowledge. The pretrained backbone resolves the named object; the action head routes position and does not read that resolution. We support this three ways—the field’s own ablations, a black-box circuit, and a null—and then show that removing the confound closes the gap at the level of the circuit.

The field’s ablations already show language is decorative. When the instruction is masked, published policies barely move: OpenVLA-OFT falls from 97% to 0.9%, π_0 from 94% to 0.6%, and $\pi_{0.5}$ from 97% to 0.0% (Fang et al., 2026c), and a vision-only controller matches a language-conditioned one (44.6% vs 47.8%) (Lian et al., 2026). A capability that can be removed with no change in behavior was never driving the policy.

A black-box circuit localizes the binding. On behavior-cloning substrates we can instrument (SmolVLA and a Qwen2.5-VL policy; we state the scope rather than claim it transfers untested), the action output moves with target position and barely with the instruction: across eight models the perturbation ratio $\Delta_{\text{instr}}/\Delta_{\text{pos}}$ is 0.03–0.12 ($\Delta_{\text{pos}} = 0.46\text{--}0.55$ vs $\Delta_{\text{instr}} = 0.017\text{--}0.023$, $n=40$ each). The object binding is nonetheless present and localizable: a single commit head (L19H17) carries it, and it is causal in one direction only. Injecting the word direction at an early layer flips the selected object 100% of the time, while steering the action read-out does nothing (0%). The binding exists upstream and the action head simply does not consult it.

A null: no training-time surrogate on the confounded data repairs it. If the knowledge were merely under-weighted, a penalty or a bottleneck on the original data would surface

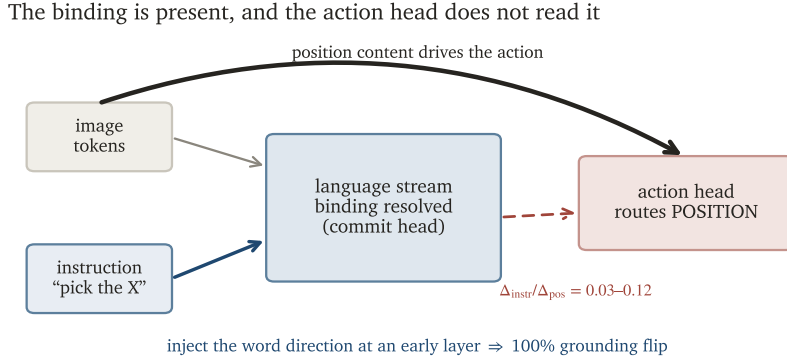


Figure 2: The binding is present and the action head does not read it. The instruction forms an object binding in the language stream (solid), but the action head is driven by position content routed from the image (heavy), and the path from the binding to the action head is weak (dashed, $\Delta_{instr}/\Delta_{pos}=0.03-0.12$). Injecting the word direction at an early layer flips the selected object 100% of the time.

it. None does: a spectral-decoupling penalty, an action-head loss reweighting, and a one-sided visual information bottleneck all sit at the disease level regardless of hyperparameter (Section 6). The route cannot be re-opened by regularizing the old data; the confound has to be removed from the data itself.

Removing the confound closes the lesion. The mechanism is causal, not merely correlational, because fixing the data fixes the circuit. In the base, the action head’s read-out is weakly aligned to the object role (cosine +0.83 on the canonical object, +0.47 on the distractor, with 22% of distractor reads anti-aligned); after the cure the same measurement reads +0.92 and +0.94 with no anti-aligned reads. The deconfounded data does not teach the backbone what the words mean—it already knew—it teaches the action head to route what the backbone resolves. This is why the cure is a data recipe and not an architecture, and it is the subject of the next section.

6 A data recipe installs grounding; no training-time surrogate does

The cure changes the data and protects the model, and nothing else. We show the two ingredients are a causal pair, and we show that four alternatives a reader would reach for first all fail.

6.1 The deconfounding principle: the instruction is the only predictor

We build demonstrations in which the instruction is the only variable that predicts the target. Every object appears as the target and as a distractor across randomized positions, and the distractors are always in frame. A position prior earns nothing, because the target is not where it was last time; a scene prior earns nothing, because every scene contains every object; only reading the name pays. The principle is one sentence, and it is the whole method. Applying it turns the base’s 13.5% into a grounded policy.

6.2 Low-rank protection is structural, not a small step

The pretrained backbone already resolves the words (Section 5), so the training must install the routing without overwriting the resolution. A low-rank update (Hu et al., 2022) does this and full fine-tuning does not: on the identical data, full fine-tuning returns 12.5% (grounding destroyed) and a rank-32 adapter returns 94.5%. The protection is structural, a property of the rank, not of the step size: doubling the learning rate barely moves the

Everything else fails: only deconfounded data $\times$ low-rank protection installs grounding

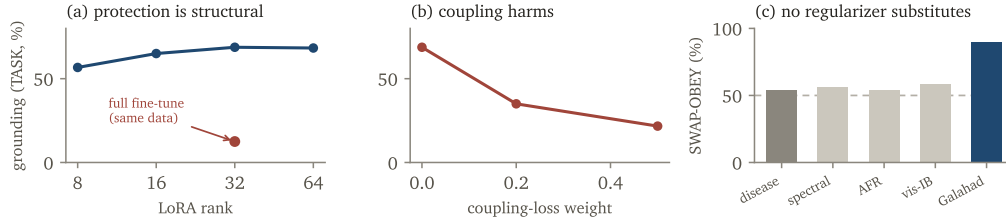


Figure 3: Only deconfounded data with low-rank protection installs grounding. (a) Protection is a property of the rank: the capacity curve rises then plateaus, and full fine-tuning on the same data collapses to 12.5%. (b) Coupling supervision harms in a dose-response. (c) Three training-time regularizers on confounded data all sit at the disease level; none approaches the deconfounded recipe.

result (68.75% to 66.7% on the development data), while releasing the rank breaks it. The protection-capacity curve rises then plateaus, 56.7 / 65.0 / 68.75 / 68.3 at ranks 8 / 16 / 32 / 64 on the development data, so rank-32 is a measured choice with no headroom above it; on clean data the curve is flatter and higher (85 at rank 8, 93 at rank 32). Low rank preserves the grounding; the deconfounded data trains the action to read it. Neither ingredient works alone.

6.3 Four alternatives fail, and the failures are the argument

A grounding objective, a training-time regularizer, and more data are the obvious substitutes, and each one is dead (Figure 3).

Coupling supervision harms monotonically. Supervising the action head’s attention to peak on the named object (Jia et al., 2026), the mechanism a reader would expect to help, degrades grounding in a clean dose-response: 68.75% with no coupling, 35% at weight 0.2, 21.7% at weight 0.5. The frozen backbone already knows where to look; forcing the action expert to be told over-constrains its read-out and breaks it.

No training-time regularizer substitutes for the data. We trained three regularizers on the original confounded data, the direct test of whether the paradigm can be fixed without the anti-shortcut dataset: a spectral-decoupling penalty on the action, an action-head loss reweighting, and a one-sided visual information bottleneck. Every arm at every hyperparameter lands at the confounded-data disease level (task 2–25%, obey 48–62%) and none approaches the deconfounded recipe (task 60%, obey 90%). A strong penalty raises a metric only by damping the motor. The regularizers cannot manufacture grounding from confounded data; the deconfounded data can.

Data scale alone is dead. On the minimal testbed, doubling deconfounded demonstrations moves object selection from 52% to 50.5%. Volume without the routing fix raises the blind floor, not the grounding.

6.4 One principle, one generator per referent type

The principle is universal and the design is per-type: color needs color orthogonal to position and shape, spatial needs side orthogonal to identity, and negation, relation, ordinal, and composition each break a different confound. All share one scripted-oracle generator and one battery, so each type is a small design, not a new method. The generator ships with the paper, and it produces the exam (the counterfactual battery) and the medicine (the training set) from the same code.

Table 1: Main battery, object substitution, all ten slots, $n=200$, on the object-family weight. The cured policy passes every face; the base has the motor and lacks the grounding.

face	question	cured	base
task	fetches the named object?	94.5% [90.6,96.8]	13.5% [8.0,21.9]
swap-obey	goes to the newly named object?	200/200	n/a
swap-obey true-tgt	avoids the trained object?	0/200	n/a
occ	needs vision?	0/20	0/20
nonsense	needs a real name?	4.5%	n/a
motor	vanilla pick-place intact?	n/a	100%

7 Experiments

We evaluate the cured policy—which we release as Galahad—on the protocol of Section 3. It grounds object identity at a rate no prior policy approaches, its released weight grounds seven referent-type axes at once, and its remaining failures are motor, not grounding.

Two weights carry the results, and we name them before any number. The object-family weight is trained on the object axis alone and holds the 94.5% of Table 1 and the prediction-face result of Section 8. The release weight folds seven referent axes into one set of weights (Table 2); it trades roughly ten points of task completion for that breadth while its grounding matches the specialist’s. Every table states which weight it was measured on.

Setup. The base is MolmoAct2-Think-LIBERO; the cure is a rank-32 low-rank adapter over both the VLM and the action expert, trained for 3,000 steps (8-way data parallelism on 24 GB consumer GPUs) on 900 deconfounded episodes—90 per object, every object appearing as target and as distractor across randomized positions. Evaluation covers all ten scene slots, and the merged model runs inference on a single 24 GB card.

Main battery. Table 1 is the object-substitution result, all ten slots, $n=200$. The policy succeeds 94.5% of the time where its base scores 13.5% and the field 0–1%. It obeys a renamed target on 200 of 200 trials and never completes the originally trained pick (0/200); it needs the pixels (blank cameras collapse it) and it needs a real name (“xxx” drops it to 4.5%). The 4.5% is a reported residual, not a suspicious zero: one scene retains a scene-to-target prior strong enough to act without a name. What moved is the reading of the instruction: the advance runs from 13.5% to 94.5% while the pick-and-place motor is untouched.

The position-prior reading is wrong, and the one cliff is a motor wall. The benchmark’s own perturbation suite refutes “it memorized where things sit.” Displacing the objects 21 cm costs nothing (96.7%), so the policy tracks objects visually. At a 35 cm displacement the score craters to 6.7%, but the reach-probe shows the arm still reaches the correct object every time (reached-target 3/3) and doubling the step budget does not rescue it: the grounding survives the displacement and the low-level grasp fails at the edge of the workspace. The cliff is a motor limit of the base, not a grounding limit of the recipe.

Three axes, three shortcuts, all cured. Object identity is one of three shortcuts the base hides. Task success plus occlusion pass on all three while swap-obey fails on all three—the point of Section 3. On the spatial suite the base scores 99% on task and 50% on swap-obey (chance for two bowls, so the 99% is a canonical-bowl shortcut); the cure raises swap-obey to 86% at a motor cost (task 51%, occlusion-clean). On the goal suite the base scores 100% on the canonical scene but 31.2% on held-out scenes with the objects displaced and the instruction fixed, reaching for the remembered position; the cure raises the held-out rate to 79.2% [65.7, 88.3] with occlusion 0/48 and the reach-probe going to the moved object 12/12.

The recipe transfers to a second simulator. On a second simulator with a different renderer and object set, the same recipe grounds the object name at swap-obey 44–60% across two seeds against a 20% chance rate, with occlusion 0. The task rate is motor-limited at 14–28%,

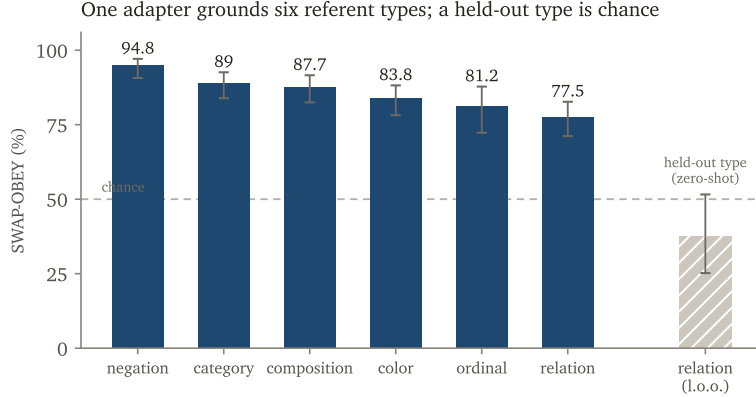


Figure 4: One adapter grounds six referent types (swap-obey 77–95%, occlusion 0 on every axis), and generality stops at its edge: a type held out of training grounds at chance. Intervals are Wilson 95%; the dashed line is chance.

Table 2: The release: seven axes on the shipped merged checkpoint. swap-obey is grounding; task is grounding times motor and is bench-dependent. Spatial’s obey count is backed by a $6.2\times$ distance separation (d_{named} median 20 mm vs d_{true} 124 mm), since its two referents are identical bowls.

axis	swap-obey	occ	task
object	99.0% (198/200; true-target 0/200)	0%	80.0%
spatial	obeyed 187 / went-true 0	0%	55.0%
goal (displaced referent)	83.3% (base 31.2%)	0%	—
color	88.9%	0%	22.2%
negation	89.6%	0%	16.7%
category	78.0%	2%	20.0%
composition	75.0% (color 87.5 / size 62.5)	0%	2.1%

the same “grounding works, motor is the ceiling” pattern; we cite the swap range rather than a point because the cross-simulator motor is seed-sensitive.

One adapter grounds six referent types. A single adapter trained across six axes at once—including two (ordinal, relation) beyond the release set—grounds all six (Figure 4; full table in Appendix D): every axis clears swap-obey 77–95% with occlusion 0, so one model reads color, category, ordinal position, spatial relation, negation, and composition. The task rates (26–51%) are the motor ceiling and swap-obey is the pure grounding metric.

Generality holds within trained coverage and stops at its edge. A model trained on five types and tested zero-shot on the sixth grounds it 37.5% of the time, at chance, against 75% when that type is in training (disjoint intervals). The edge is the type, not composition: the same adapter zero-shots a never-trained composition of two trained types—“the second red one from the left”—at 64.2% where its adapter-ablated base scores 16.1% (paired +55.6 points, McNemar $p \approx 10^{-5}$). Trained types compose; an untrained type does not transfer.

The release grounds seven axes in one weight. The released artifact folds seven axes into one merged checkpoint, and Table 2 is measured on that shipped weight. Every axis grounds (swap-obey 75–99%) and every axis is vision-gated (occlusion $\leq 2\%$). On object identity the release obeys the renamed target 198 of 200 times and never fetches the trained object, matching its adapter, while completion reads 80.0% against the specialist’s 94.5%: the gap is motor (reach-to-true-target 90.5%), the measured price of carrying seven axes in one set of weights, and grounding pays none of it.

The task column is bench-dependent in both directions. The low task rates on the second simulator’s axes are the bench: the single-axis anchors on the same bench read task 26–51% against swap-obey 77–95%, and the release sits 10–15 points under those task anchors—the same unification price as the object axis—while its grounding sits at or above them. Spatial inverts the other way: the uncured base scores 99% on task and exactly chance on swap-obey, so task alone would reverse the conclusion, and the release is the one reading the spatial phrase. Composition decomposes cleanly: the color attribute follows at 87.5% and size at 62.5%, so the short leg is size, which is actionable where a pooled 75% is not.

8 A grounded prediction face

The same deconfounding principle applies to a second output. On the object family we train the object-family weight with a foresight branch that predicts the task-relevant future region, and we test that prediction with the counterfactual of Section 3: holding the pixels fixed and changing one word, the predicted region of motion moves to the newly named object 58.3% of the time against a 25% chance rate ($p < 10^{-6}$, $n=48$), and blanking the cameras kills the effect (0/48), so the prediction reads the pixels rather than a name-to-position table (Appendix B). This is one set of weights whose action and whose prediction of the future both obey the instruction—the low-rank update folds into the base and the foresight head lives in the same checkpoint, the fold changing the action by a bounded 1.17×10^{-2} . Across the world-action models of Section 4 the predicted future does not obey the instruction, so we read this as a verifiably grounded prediction face rather than a runtime imagination: it is not action-conditioned and not rollable, and foresight branches of this kind are routinely removable at inference (Fang et al., 2026b). The two faces are not equally grounded, and we report the asymmetry rather than borrow one number for the other: the action face obeys at 100% and the prediction face switches at 58.3%. Adding the prediction face costs no action accuracy—trained jointly, the two-branch model ties the action-only control (93.3% each, gap 0.0 points), and on a non-saturated bench the action-only, two-branch, and read-back-masked variants read task 48/53/47% and grounding 87/88/86% with fully overlapping intervals—so the prediction face is grounded and neither helps nor taxes the action.

9 Analysis and boundaries

The result is a policy that grounds language by data, and its edges are as informative as its center. We state four as findings.

Data is the active ingredient; a training-time regularizer is not. A penalty, a reweighting, and a bottleneck on the confounded data all fail (Section 6); grounding comes only from making the instruction the sole predictor of the target. The deconfounded data is the drug and the low-rank update is the protection that lets the pretrained resolution survive—neither works alone.

Grounding costs appearance, not a demonstration. Grounding follows an exposure gradient (Figure 5): an object trained as a target grounds at 78%, an object seen only as a distractor grounds at 61%, and an object never in the data grounds at 0%. The marginal cost of a new object’s grounding is that it appear in a scene, not that it be grasped—the recipe scales by coverage at low marginal cost.

The vocabulary is closed. An object removed from training entirely, target and distractor, grounds at 0 of 16; told its name, the arm goes to a trained object instead, while trained-object controls hold at 75%. The recipe grounds a closed vocabulary and does not extend to a never-seen word; open-vocabulary grounding is a different problem we do not claim.

Grounding trades against shortcut robustness by design. On a perturbation benchmark whose Language factor paraphrases the instruction and rewards a policy whose actions do not change, the cured model scores below its base: 76.4% pooled against 90.0%, with the Language factor falling 96.0% to 72.0% (−24 points). A policy that ignores language is

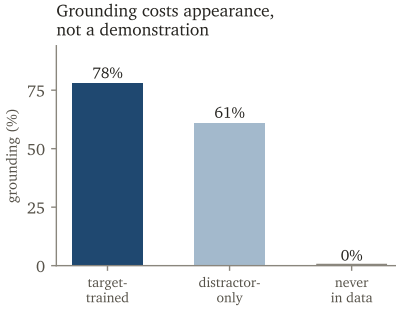


Figure 5: The scaling economics of the recipe: an object needs only to appear in training scenes for its grounding to become learnable; no grasp demonstration of it is required. Target-trained 78%, distractor-only 61%, never-in-data 0%.

paraphrase-invariant and scores high there; ours reads the instruction and scores lower. The Language axis measures language-invariance, not grounding, and the 24-point gap is the measured price of reading the words, not a regression. We never report this benchmark’s aggregate as our robustness.

The cure needs a base that can route. The same recipe on a second base is a null with a mechanism. On $\pi_{0.5}$, the low-rank cure under both adapter placements (action expert only, and VLM plus expert) leaves swap-obey at chance (53.3% against a 46.7% base, $n=60$ each); on a frontier omnimodal world model’s action head the cure is a null as well (a pilot, $n=12$). The shared factor is architectural: both condition their action expert on a pooled summary of the backbone, so the token-level binding the recipe protects is never exposed to the action head, and there is nothing to route. MolmoAct2’s per-layer interleaving exposes the binding to the action head, and that exposure is what the cure routes through. The recipe is a data cure conditioned on a routable architecture, not a universal patch—which sharpens the diagnosis: instruction blindness is a severed route, and the cure re-opens it only where a route exists to re-open.

10 Conclusion

Instruction blindness—a VLA policy that passes an instruction benchmark without reading the instruction—is a concrete instance of causal confusion, and a change-one-word counterfactual measures it: on object identity the published field selects the correct object 0–1% of the time. The cause is a severed route, not missing knowledge; the pretrained backbone resolves the named object and the action head routes position and never reads that binding. Because the knowledge is present, the repair is a data recipe rather than an architecture: making the instruction the only predictor of the target and protecting the pretrained resolution with a low-rank update installs grounding where full fine-tuning on the same data destroys it. The cured policy grounds seven referent-type axes in one set of weights, transfers to a second simulator, and grounds its predicted future as well as its action, and its boundaries are findings—coverage-bound, closed-vocabulary, and conditioned on a base whose backbone grounding is exposed to the action head. We release the policy, the deconfounded data and its generator, and the measurement code.

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A The conditioning gradient of the predicted future

Grounding of a predicted future tracks how the instruction reaches the prediction, not model scale (Figure 6). A world model whose future is conditioned on the action cannot read the instruction at the frame level—the name reaches the pixels only through a weak action head, and such models hallucinate a successful future even under a wrong action (Liu et al., 2026). A world model whose future is conditioned on the text reads the name directly. We ran the switch counterfactual on a text-conditioned video world model (NVIDIA, 2025a) over held-out frames, hand-scored in pixels under a continuity protocol that rejects composited outcomes: it obeys the swapped object name 64.7% of the time [41.3, 82.7] against a 25% chance rate, the strongest grounding in the world-model zoo. The tier replicates at the frontier: Cosmos 3’s Nano model (NVIDIA, 2026), two generations newer, lands at 77.8% [61.9, 88.3] on the same protocol. The gradient is one disease along one spectrum: text-conditioned generation grounds (64.7–77.8%), action-conditioned world models do not (20–45%), and imitation policies fail (0–1%).

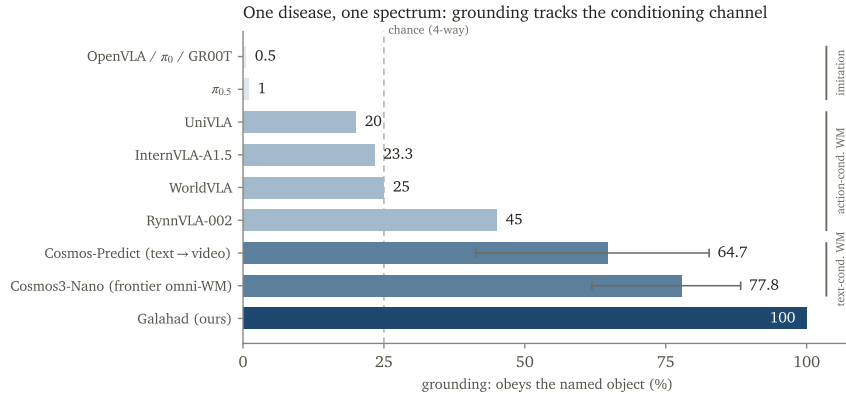


Figure 6: One disease, one spectrum. Grounding of the output tracks how the instruction reaches it: text-conditioned generation reads the name, action-conditioned world models bottleneck it through a weak action head, and imitation policies discard it. Obey rate against a 25% chance rate.

B The prediction face, illustrated

Figure 7 illustrates the two grounded outputs of Section 8: changing one word moves both the action endpoint and the predicted region of motion to the newly named object.

Table 3: The prediction face, object-family weight, $n=48$. The predicted future obeys the instruction and needs the pixels; the action face is more grounded than the prediction face.

counterfactual	result	chance
switch (one word, predicted region moves to named object)	58.3% ($p < 10^{-6}$)	25%
blind switch (same, cameras blank)	0/48	25%
action face swap-obey (Section 7)	100%	n/a

C Real-robot verification

The battery’s load-bearing control transfers to hardware. We ran the same recipe on a low-cost SO-101 arm (commodity servos, a single third-person camera) and verified the vision-gating claim directly: with the cameras blanked the policy collapses to a single instruction-independent attractor, and with vision on the reach varies with the named color. On the

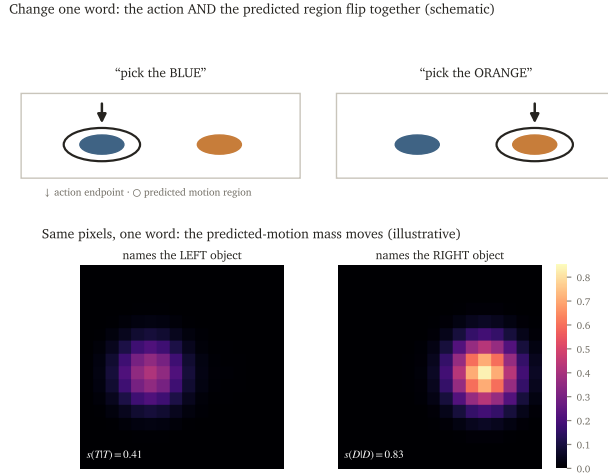


Figure 7: Two faces of one weight. Top: changing one word moves both the action target and the predicted region of motion to the newly named object (schematic). Bottom: the predicted motion mass over two objects, same pixels, one word changed; the mass moves two orders of magnitude (illustrative of the measured $s(T|T)=0.41$, $s(D|D)=0.83$).

physical arm, reach-obey is 10/21 (48% [28, 68], chance 33%) and grasp-obey is 5/15; the reach-to-grasp gap is actuation (the SO-101 is mechanically marginal), attributed by the reach-probe. We adjudicate by pixel co-location in a side camera, never by joint angles, and we make no significance claim: the interval includes chance and $n < 48$, so hardware is a qualitative confirmation of vision-gating, not a second quantitative result. We do not report a per-color story, whose per-cell intervals all cross chance.

D The referent-type axes in full, and the verb axis

Table 4 is the full per-axis table for the single six-axis adapter of Section 7 (visualized in Figure 4).

Table 4: One adapter, six referent-type axes, $n \geq 96$ per axis. swap-obey is grounding; task is grounding times motor. Leave-one-out: a type held out of training grounds at chance (37.5% vs 75% trained, disjoint intervals).

axis	category	color	ordinal	relation	negation	composition
swap-obey	89.0	83.8	81.2	77.5	94.8	87.7
occ	0	0	0	0	0	0

An eighth referent type grounds the verb. On a deconfounded lift-versus-slide design (the same object under both verbs), the cured policy’s verb selectivity is +44.3 points: told to lift, it lifts 46.3% of the time; told to slide, it lifts 2.0% (disjoint intervals); the base’s selectivity is +0.0. The lift program collapses to zero with the cameras blanked, and the render shows a real grasp-then-raise against a real push-along-surface. The magnitude is motor-limited (pooled obey 37%, a grasp-stability wall), and the claim is the selectivity: the word chooses which motor program runs. Joint training dilutes this axis (+6 points in the unified weight), so the released checkpoint does not carry it and we report the single-axis result.

E Measurement discipline

Every headline in this paper survived an attempt to disconfirm it, and four times the attempt succeeded and changed the number. We record the four, because a measurement is only as trustworthy as its authors’ willingness to catch themselves. Our standing rule is that a 100% or a 0% is a suspect, not a trophy, until a disconfirming control rules out the instrument, and that we verify in the modality the truth lives in—rendering the rollout and looking at the pixels rather than trusting a scalar.

Case 1: the probe that verified grounding was ill-posed. An early instruction-swap probe fed each task the next task’s instruction verbatim, and on most slots the named object was not in the scene, so a zero was uninterpretable. We rebuilt it as swap-obey, drawing the decoy from objects actually present; every grounding number from the ill-posed probe was void.

Case 2: a structural zero in the data masqueraded as a policy limit. Two slots scored zero for weeks and were filed as a motor limit. A two-episode trace of the demonstration generator found the scripted oracle rammed a tall bottle at hover height and closed on air, so those slots had no demonstrations to learn from. The fix was in the generator and the headline moved 22 points with no change to the method.

Case 3: our own reported effect did not replicate. We reported that the two-branch model taxed grounding by 21.6 points. A second seed erased it: the tax lived inside single-seed variance and its sign flipped across seeds. We retract the tax and report the null (Section 8).

Case 4: a metric can lie by deflating. A color axis read task 0% and looked like an honest motor failure. It was a scorer aggregation bug; the corrected rate is 37.5% and the grounding was intact the whole time (swap-obey 83.8%). We now verify a suspiciously low number as hard as a suspiciously high one. Separately, the success predicate itself can be gamed: an object that spawns inside the goal region scores a do-nothing success, so we lift-gate success and measured the init value of every metric.

F Referent-type generator designs

One principle—the instruction is the only predictor of the target—instantiates as one small design per type, sharing a scripted-oracle generator and the battery (Table 5).

Table 5: One generator, one design per referent type: the confound each breaks and the counterfactual that verifies it.

type	confound broken	swap counterfactual
object identity	position, scene	rename the target object
color	color $\perp$ position $\perp$ shape	change the color word
category	category $\perp$ instance $\perp$ co-appearance	change the category word
ordinal	ordinal $\perp$ absolute position $\perp$ identity	change the ordinal word
spatial relation	side $\perp$ identity	name the other referent
negation	keyword-match	flip the negated attribute
composition	single-attribute lookup	change one attribute in the pair
goal	remembered destination	displace objects, hold the goal word

G Compute and release

All models train and run on commodity GPUs. Each Galahad variant trains with 8-way data parallelism on NVIDIA RTX 3090s (24 GB each), and the merged model runs inference on a single 24 GB card; the one exception is the 14B text-to-video world model of Appendix A, which we ran on an A100. We release four artifacts: the policy (the seven-axis merged weight and the object-family dual-output weight, rebuilt from the released recipe and battery-verified), the deconfounded referent-type datasets in a standard format, the generator that

produces both the exam and the training set from one principle, and the one-command measurement battery that reports swap-obey, occ, nonsense, and the reach-probe for any policy. Per-slot and per-object tables with Wilson intervals, seed ranges, the protection-capacity and coupling-dose curves, the regularizer-null sweep, and the LIBERO-Plus per-factor table ship with the code alongside the frozen per-slot result artifacts.